

Programming for AI Lab Tasks

Lab#2

1. We are creating a database where we are going to store following items

Customer Orders

- * order_id
- * customer_name
- * order_items
- * discount
- * order_total
- * order_date

Temperature Monitoring

- * sensor_id
- * location
- * temperature_readings (list)
- * average_temperature
- * timestamp

Your job is to identify the data-type to be used for each of above values. Also create an example for one record using variables and values of appropriate data-types. Print Values

2. Announcement Preview

You're developing an announcement preview tool for a messaging platform. The tool needs to provide a concise preview of longer messages.

Task:

Given a message: "Join us for the webinar on Aug 30, 2023, 2:00 PM (GMT)", perform the following tasks:

- * Calculate the length of the message.
- * Replace the word "webinar" with "online session".
- * Extract the date "Aug 30, 2023".

- * Create a shortened preview containing only the first 40 characters.
- * Extract time.

Print all the results.

3. Fitness Progress Tracker

You're developing a fitness progress tracker that records different exercises performed each day.

Task:

Given exercise logs: ['Running', 'Push-ups', 'Yoga', 'Swimming'], perform the following tasks:

- * Create an empty log for cardio exercises.
- * Append 'Cycling' to the cardio log.
- * Insert 'Plank' as the second exercise in the log.
- * Extract the exercise 'Yoga' from the list.
- * Create a brief cardio log containing only the first two exercises.
- * Extract all items from list
- * Extract all items except first and last from list

Print all the results.

4. Bookshelf Organizer

You're working on a bookshelf organizer for a library app.

Task:

Given a list of book titles: ['Book A', 'Book B', 'Book C', 'Book D'], perform the following tasks:

- * Create an empty list for recommended books.
- * Append 'Recommended Book' to the recommended books list.
- * Insert 'Bestseller' as the second book in the list.
- * Extract the last book from the list.
- * Create a concise book list containing only the first three books.
- * Insert "Thore" as last item in list

Print all the results.

5. Coulomb's Law Calculator

You're developing a calculator to apply Coulomb's law for electrostatic interactions.

Scenario:

As a user, you'll input the charges (q_1 and q_2) of two objects and the distance (r) between them. The calculator will determine the electrostatic force (F) between the objects.

Task:

- * Enter the charges of the two objects (q_1 and q_2) and the distance (r).
- * Calculate the electrostatic force (F) using Coulomb's law: $F = k * q_1 * q_2 / r^2$, where k is the electrostatic constant.